

Harshit Kumar Singh

ML Engineer | System Administrator

Contact : 7800333757 | harshitsingh097@gmail.com
Address : Pune, Maharashtra
GitHub : <https://github.com/HARSHIT097>
LinkedIn : <https://www.linkedin.com/in/harshitsingh097/>

Professional Summary

Results-driven professional with a Master's in Computer Applications and a Postgraduate Diploma in Artificial Intelligence, I bring over 4 years of hands-on experience in Python programming, along with a solid foundation in predictive and statistical analysis. I have a proven ability to design and implement advanced neural network models using frameworks such as TensorFlow and PyTorch. Leveraging strong project management and communication skills, I have successfully led and contributed to diverse end-to-end AI projects, including time-series forecasting, computer vision, and chatbot development. My goal is to continuously learn and apply cutting-edge AI techniques to solve real-world challenges while driving innovation and organizational growth.

Professional Experience

Tata Consultancy Services (TCS) | System Engineer

Pune, Maharashtra, India | Oct 2021 – Present | 3+ years

- Oversaw and maintained critical IT infrastructure, ensuring system availability and promptly resolving technical issues.
- Developed the NAMSA Dashboard Tool using Python scripting and automation techniques to streamline troubleshooting processes.
- Automated repetitive tasks, significantly reducing manual workload and enhancing overall system performance.
- Managed daily Business-as-Usual (BAU) activities, Linux system administration, and SQL-based data handling.
- Continuously upskilled in emerging technologies to drive seamless operations and improve IT infrastructure performance.

Abroad Shiksha | Associate Product Engineer

Gurugram, India | Jan 2017 – July 2017 | 0.7 years

- Led the development and optimization of internal tools to support product engineering initiatives, focusing on system reliability and performance.
- Designed and deployed a chatbot using **IBM Watson Assistant** and **Python**, significantly enhancing user support and reducing repetitive queries by over **40%**.
- Implemented intelligent automation scripts to streamline workflows, improve response time, and minimize manual intervention across the product lifecycle.
- Monitored and analyzed key performance metrics, continuously improving chatbot accuracy and system efficiency through data-driven iterations.
- Collaborated cross-functionally to manage Business-as-Usual (BAU) operations, performed **Linux system administration**, and utilized **SQL** for data insights and troubleshooting.
- Actively explored and integrated emerging technologies to enhance user experience and product scalability.

Personal Projects

NAMSA Tool | 2024 - Current

- Designed and developed a dashboard leveraging Python scripting for streamlined troubleshooting and efficient data access.

Stock Price Prediction using LSTM & GRU | 2023

- Built a time-series prediction model to forecast stock prices using LSTM and GRU neural networks.
- Processed datasets with daily and minute-wise records, leveraging APIs for real-time updates and creating dynamic visualizations.
- Delivered the solution within a tight deadline and deployed an interactive Streamlit interface for user-friendly access.

SMEEP: Employee Efficiency and Productivity Tool| 2022

- Created an end-to-end cross-platform software to monitor employee activities across six functional modules.
- Implemented face recognition using OpenCV, background process tracking, and activity screenshots for comprehensive monitoring.
- Deployed the application using PyQt5 for the interface, REST APIs, and AWS servers, ensuring scalability and real-time performance.

Athena Chatbot | 2019 - 2021

Developed a virtual assistant powered by IBM Watson to address domain-specific queries, schedule meetings, and create counselling sessions

Technical Skills

Programming Languages	: Python, SQL
Operating System / Platforms	: Windows, Linux
Data Visualization	: Tableau, Power BI, Matplotlib, Seaborn, Excel
Machine Learning	: Data Analysis, Data Visualization, Predictive & Statistical Modeling, Semantic and Instance Based Analysis
Deep Learning & Machine Vision	: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, GANs
Web Frameworks	: Django, Flask
Databases	: MySQL, Hadoop, PySpark
Web Development	: HTML, CSS, JavaScript

Education

- PG Diploma | CDAC ACTS, Pune | 2021 | Specialized in AI & ML with real-world applications | GPA: 6.8
- Master's in computer applications (MCA) | GLA University, Mathura | 2019 | GPA: 8.7
- Bachelor's in computer applications (BCA) | Subharti University, Meerut | 2017 | GPA: 6.9

Certifications & Achievements

- **AI4CPS Certificate Programme**
IIT Kharagpur | 2024 | Hands-on AI for Cyber-Physical Systems.
- **AWS Certified Solutions Architect - Associate**
AWS | 2024 | Expertise in scalable systems and cloud